

Evaluation Phase IV

1 Overview

The following list provides an overview of the WCPs we used for the Phase I of the work:

- WCP-1 Sequence
- WCP-2 Parallel Split
- WCP-3 Synchronization
- WCP-4 Exclusive Choice
- WCP-5 Simple Merge
- WCP-6 Multi-Choice
- WCP-7 Structured Synchronizing Merge
- WCP-10 Arbitrary Cycles
- WCP-17 Interleaved Parallel Routing
- WCP-19 Cancel Activity

The goal of this part of the evaluation is to show the generalizability of the developed algorithms. By using the examples from Phase I which failed, we already showed the validity of the algorithms. So the goal is not to cover a complete set of process structures, but instead to implement change operations for unseen process structures and show, that the algorithms also work correctly for them.

With this goal, we define the following 5 combinations of WCPs for which we define change operations. The goal is that for each of the three failure conditions, activities happening in between, contradictions between inputs and violation of locked dependencies; per newly defined process, we have one case to consider, if applicable.

We define the following new process models, build up from combinations of WCP, to be used:

1. Simple merge nested with synchronization
2. Parallel split nested with exclusive choice
3. Synchronization followed by exclusive choice
4. Interleaved parallel routing with multi choice
5. Exclusive choice followed by exclusive choice

2 Results

In this section we present the results of the evaluation for the 5 cases of combined WCPs to create unseen process structures.

2.1 Simple merge nested with parallel split

Simple merge: AD, BD, CD

Parallel split: XYZ, XZY, YXZ, YZX, ZYX, ZXY

For the combination, one of the branches in the simple merge becomes the parallel split, here we choose to replace activity A with the synchronization

Acceptance sequences: XYZD, XZYD, YXZD, YZXD, ZYXD, ZXYD, BD, CD

- Activities happening in between (parallelize / collapse) Collapse $\{X, Y, D\}$ with new activity name H

	C	B	Z	H
C	.	$(-, \bar{\wedge})$	$(-, \bar{\wedge})$	$(\prec_d, \Rightarrow)$
B	$(-, \bar{\wedge})$	.	$(-, \bar{\wedge})$	$(\prec_d, \Rightarrow)$
Z	$(-, \bar{\wedge})$	$(-, \bar{\wedge})$	.	$(\prec_d, \Rightarrow)$
H	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$	.

Table 1: Activity relationships matrix for activities C, B, Z, H ; move all to activity D

- Contradictions between inputs Move activity D with $(D \Leftrightarrow B), (D \Leftrightarrow X)$

	C	X	D	Y	Z	B
C	.	$(-, \Rightarrow)$	$(\prec, \Rightarrow)$	$(-, \bar{\wedge})$	$(-, \bar{\wedge})$	$(\prec, \Rightarrow)$
X	$(-, \Leftarrow)$	.	$(\prec, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(\prec, \Leftarrow)$
D	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	.	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	$(-, \Leftarrow)$
Y	$(-, \bar{\wedge})$	$(-, \Rightarrow)$	$(\prec, \Rightarrow)$	.	$(-, \Leftarrow)$	$(\prec, \Rightarrow)$
Z	$(-, \bar{\wedge})$	$(-, \Rightarrow)$	$(\prec, \Rightarrow)$	$(-, \Leftarrow)$	.	$(\prec, \Rightarrow)$
B	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	.

Table 2: Activity relationships matrix for activities C, X, D, Y, Z, B ; combined similarity score

- Violation of locked dependencies Lock $(X \Leftrightarrow Y)$ Move X with the dependency input $(X \Leftrightarrow D)$

	X	B	C	Y	Z	D
X	.	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
B	$(-, \Rightarrow)$	.	$(-, \bar{\wedge})$	$(-, \Rightarrow)$	$(-, \bar{\wedge})$	$(\prec, \Rightarrow)$
C	$(-, \Rightarrow)$	$(-, \bar{\wedge})$	.	$(-, \Rightarrow)$	$(-, \bar{\wedge})$	$(\prec, \Rightarrow)$
Y	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$	.	$(-, \Leftarrow)$	$(\prec, \Leftarrow)$
Z	$(-, \Rightarrow)$	$(-, \bar{\wedge})$	$(-, \bar{\wedge})$	$(-, \Rightarrow)$	.	$(\prec, \Rightarrow)$
D	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	.

Table 3: Activity relationships matrix for activities X, B, C, Y, Z, D

2.2 Parallel split nested with exclusive choice

Exclusive choice: A, B

Parallel split: XYZ, XZY, YXZ, YZX, ZYX, ZXY

For the combination, one of the branches in the synchronization becomes the exclusive choice, here we choose to replace activity X with the exclusive choice

Acceptance sequences: AYZ, AZY, YAZ, YZA, ZYA, ZAY, BYZ, BZY, YBZ, YZB, ZYB, ZBY

- Activities happening in between (parallelize / collapse) Not applicable here, since no temporal ordering of activities exists
- Contradictions between inputs Insert new activity H with $(H \Leftrightarrow Y), (H \Leftrightarrow A)$

	A	H	Z	Y	B
A	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$
H	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$(-, \Leftarrow)$
Z	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$
Y	$(-, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(-, \Leftarrow)$
B	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	.

Table 4: Activity relationships matrix for activities A, H, Z, Y, B

- Violation of locked dependencies Lock $(A \not\Leftarrow B)$ Parallelize A, B

	Z	A	B	Y
Z	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$
Y	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.

Table 5: Activity relationships matrix for activities Z, A, B, Y ; resolve by deleting locked dependencies violated

2.3 Synchronization followed by exclusive choice

Exclusive choice: A, B, C

Synchronization: XYZQ, XZYQ, YXZQ, YZXQ, ZYXQ, ZXYQ

For the combination, the activity for synchronization is replaced by the the exclusive choice.

Acceptance sequences: XYZA, XZYA, YXZA, YZXA, ZYXA, ZXYA, XYZB, XZYB, YXZB, YZXB, ZYXB, ZXYB, XYZC, XZYC, YXZC, YZXC, ZYXC, ZXYC

- Activities happening in between (parallelize / collapse) Parallelize $\{X, A\}$

	A	Y	Z	X	B	C
A	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
Y	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
Z	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
X	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(-, \Leftarrow)$	$(-, \Leftarrow)$
B	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	.	$(-, \overline{\wedge})$
C	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \overline{\wedge})$	.

Table 6: Activity relationships matrix for activities A, Y, Z, X, B, C; Parallelize including Z, Y

- Contradictions between inputs Modify $(X \Leftrightarrow A)$

	X	B	Y	C	A	Z
X	.	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(\succ, \Rightarrow)$	.	$(\succ, \Rightarrow)$	$(-, \overline{\wedge})$	$(\succ, \Rightarrow)$	$(\succ, \Rightarrow)$
Y	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	.	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
C	$(\succ, \Rightarrow)$	$(-, \overline{\wedge})$	$(\succ, \Rightarrow)$	.	$(\succ, \Rightarrow)$	$(\succ, \Rightarrow)$
A	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	.	$(-, \Leftrightarrow)$
Z	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	.

Table 7: Activity relationships matrix for activities X, B, Y, C, A, Z; combined similarity score

- Violation of locked dependencies Lock $(X \Leftrightarrow Y)$ Move $X \Leftrightarrow A$

	A	C	Y	B	Z	X
A	.	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
C	$(\succ, \Rightarrow)$	.	$(\succ, \Rightarrow)$	$(-, \overline{\wedge})$	$(\succ, \Rightarrow)$	$(-, \Rightarrow)$
Y	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	.	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(\succ, \Rightarrow)$	$(-, \overline{\wedge})$	$(\succ, \Rightarrow)$	.	$(\succ, \Rightarrow)$	$(-, \Rightarrow)$
Z	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftarrow)$	.	$(-, \Leftrightarrow)$
X	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$(-, \Leftrightarrow)$	.

Table 8: Activity relationships matrix for activities A, C, Y, B, Z, X; combined occurrence similarity

2.4 Interleaved parallel routing with multi choice

Interleaved parallel routing: ABC, ACB, CAB

Multi choice: X, Y, XY, YX

For the combination, the parallel routed activity becomes a multi choice.

Acceptance sequences: ABX, AXB, XAB, ABY, AYB, YAB ABXY, AXYB, XYAB, ABYX, AYXB, YXAB

- Activities happening in between (parallelize / collapse) Not applicable here; only A and B are temporally dependent; so no ordering with activities happening in between
- Contradictions between inputs Modify ($X \not\Rightarrow A$)

	Y	X	B	A
Y	.	$(-, \vee)$	$(-, \Rightarrow)$	$(-, \Leftarrow)$
X	$(-, \vee)$	.	$(-, \Rightarrow)$	$(-, \not\Rightarrow)$
B	$(-, \Leftarrow)$	$(-, \Leftarrow)$	.	$(\succ, \Leftarrow)$
A	$(-, \Rightarrow)$	$(-, \not\Rightarrow)$	$(\prec, \Rightarrow)$	.

Table 9: Activity relationships matrix for activities Y, X, B, A ; combined similarity score

- Violation of locked dependencies Lock ($Y \Rightarrow A$) Parallelize $\{X, Y\}$

	X	A	B	Y
X	.	$(-, \Rightarrow)$	$(-, \Rightarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Leftarrow)$	.	$(\prec, \Leftrightarrow)$	$(-, \Leftarrow)$
B	$(-, \Leftarrow)$	$(\succ, \Leftrightarrow)$	.	$(-, \Leftarrow)$
Y	$(-, \Leftrightarrow)$	$(-, \Rightarrow)$	$(-, \Rightarrow)$	.

Table 10: Activity relationships matrix for activities X, A, B, Y ; combined similarity score

2.5 Exclusive choice followed by exclusive choice

Exclusive choice: A, B, C

Synchronization: X, Y, Z

For the combination, the exclusive choice is followed by an exclusive choice

Acceptance sequences: AX, AY, AZ, BX, BY, BZ, CX, CY, CZ

- Activities happening in between (parallelize / collapse) not applicable here
- Contradictions between inputs Modify ($X \prec A$)

	X	C	Y	Z	A	B
X	$\cdot$	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\succ_d, -)$
C	$(\prec_d, -)$	$\cdot$	$(\prec_d, -)$	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$(-, \overline{\Lambda})$
Y	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$\cdot$	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$(\succ_d, -)$
Z	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$\cdot$	$(\succ_d, -)$	$(\succ_d, -)$
A	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\prec_d, -)$	$\cdot$	$(-, \overline{\Lambda})$
B	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$\cdot$

Table 11: Activity relationships matrix for activities X, C, Y, Z, A, B ; combined similarity score

- Violation of locked dependencies Lock ($A \prec X$) Swap A with Y

	X	C	Y	Z	A	B
X	$\cdot$	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\succ_d, -)$
C	$(\prec_d, -)$	$\cdot$	$(\prec_d, -)$	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$(-, \overline{\Lambda})$
Y	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$\cdot$	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$(\succ_d, -)$
Z	$(-, \overline{\Lambda})$	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$\cdot$	$(\succ_d, -)$	$(\succ_d, -)$
A	$(\succ_d, -)$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\prec_d, -)$	$\cdot$	$(-, \overline{\Lambda})$
B	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$(\prec_d, -)$	$(\prec_d, -)$	$(-, \overline{\Lambda})$	$\cdot$

Table 12: Activity relationships matrix for activities X, C, Y, Z, A, B ; combined similarity score